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Work package number	WP01 – Project Coordination and Management
Work package leader	P01 – Technical University of Munich (TUM)
Abstract:	This initial version of the ProPollSoil Data Management Plan (DMP) provides a strategy for producing, storing, curating, and sharing project data in compliance with FAIR principles (findability, accessibility, interoperability, and reusability). As the project progresses, the DMP will be improved and adapted to the project's needs within D1.3 (updated version) and D1.4 (final version).
Keyword List:	Data management, FAIR data, data processing, data storage, data reuse, data review, data security

*names are ordered alphabetically

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Executive Summary

Purpose

The ProPollSoil Data Management Plan (DMP) sets out a comprehensive strategy for producing, storing, curating, and sharing project data in compliance with FAIR principles (findability, accessibility, interoperability, and reusability). The ProPollSoil project will generate a wide range of data types — including soil properties, pollinator records, ecological interactions, genomic sequences, geospatial layers, socioeconomic surveys, images, videos, and modelling outputs. This DMP is a "living" document, for use primarily by the consortium, that will be improved and updated as the project progresses, with three versions published and openly shared (at months 6, 24 and 46 of the project).

Intended audience

The intended audience of DMP is the consortium members of the ProPollSoil project, including researchers, students, data managers, and technical staff across all participating institutions.

Description of the main activities

In Section 2, the plan describes the wide range of data types expected, amounting to an estimated 75–100 TB. These datasets will be produced using standardised protocols and stored in open, interoperable formats, with secure storage and internal exchange supported through institutional servers and a dedicated LMU-hosted Nextcloud environment with controlled access, versioning, and redundancy. Section 2 also defines a structured dataset review system, supported by a central Dataset Registry, to keep track of dataset quality, metadata completeness, reproducibility, and alignment with FAIR requirements. Only reviewed datasets may feed into project outputs, and all data and code must be shared internally prior to submission.

The strategy for compliance with FAIR principles is detailed in Section 3. It mandates the use of persistent identifiers, deposition in Zenodo and domain-specific repositories, and the application of community standards such as Darwin Core, EML, ISO 19115-1:2014/Amd 2:2020 and Dublin Core. Clear rules govern licensing, embargoes, and access restrictions for sensitive or confidential data, while we aim to maintain interoperability through open formats, controlled vocabularies, and semantic links.

In Section 4, we describe the management of physical samples — soil, pollinator specimens, and plant material — including long-term preservation in institutional collections, as well as the handling of modelling outputs and software. Section 5 explains how resource needs for FAIR data stewardship are integrated into the project budget.

Data security measures in Section 6 include controlled access, secure storage, regular backups and anonymisation where required. Ethical considerations in Section 7 cover biological sampling permits, GDPR-compliant handling of human data, and responsible use of AI. Finally, in Section 8 we clarify data ownership, open-access expectations, and intellectual property provisions in line with the ProPollSoil Grant Agreement.

Research and practice implications

By ensuring that all datasets are systematically documented, assigned persistent identifiers, and made accessible through interoperable formats and standardised vocabularies, the DMP provides guidelines for high-quality, reproducible research. This approach enables researchers within and beyond the consortium to efficiently discover, integrate, and reuse data across disciplines, fostering collaboration and accelerating scientific progress in understanding the interactions between soil health,

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management practices, and soil-dependent pollinators, including their reciprocal effects on soil systems. The explicit documentation of data provenance, including links between raw data, processed outputs, and computational workflows, further enhances transparency and trust in the results, supporting advanced analyses and the development of innovative methodologies.

Conclusion

This document describes the activities conducted under the ProPollSoil project to provide clear instructions for FAIR data principles and on how to collect, generate, process and store datasets within the project and on how to make them publicly accessible. The project activities will be implemented in line with the Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data.

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List of Tables

Table 1: Non-exhaustive list of identified trustworthy and established repositories, registries and prioritized data types in addition to general ones like Zenodo and FigShare as well as registration with EuroAPPA..... 13

Terms and abbreviations

ARK	Archival Resource Key
BOLD	Barcode of Life Data System
CSA	Case Study Area
COI	Cytochrome Oxidase I
DC	Dublin Core
DCMI	Dublin Core Metadata Initiative
DMB	Data Management Board
DMP	Data Management Plan
DNA	Deoxyribonucleic acid
DOI	Digital Object Identifier
DoPI	Database of Pollinator Interactions
DwC / DwC-A	Darwin Core / Darwin Core Archive
EML	Ecological Metadata Language
ENA	European Nucleotide Archive
EuroAPPA	European Atlas of Plant Pollinator Associations
ESDAC	European Soil Data Centre
EUSO	European Soil Observatory
FAIR	Findable, Accessible, Interoperable, Reusable
GBIF	Global Biodiversity Information Facility
GDPR	General Data Protection Regulation
GloBI	Global Biotic Interactions
ITS	Internal transcribed spacer of the rRNA
IP	Intellectual Property
IPR	Intellectual Property Rights
ISO	International Organization for Standardization
LRZ	Leibniz Supercomputing Center
NCBI	National Center for Biotechnology Information
OA	Open Access

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OBO	Open Biological and Biomedical Ontologies
rRNA	Ribosomal ribonucleic acid
UTF-8	Unicode Transformation Format - 8-bit
WP	Work Package

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1 Introduction

1.1 Project abstract

ProPollSoil is a pan-European project that investigates how soil health and management practices affect soil-dependent pollinators, and how these pollinators in turn affect soil health. It combines multidisciplinary approaches, including soil science, pollinator ecology, biodiversity monitoring, modelling, and socioeconomic analysis, to identify threats and develop pollinator-safe soil management strategies. The project supports the EU Mission "A Soil Deal for Europe" and the Pollinators Initiative by improving monitoring methods, reducing soil pollution, and promoting soil literacy. Through collaboration with Case Study Areas and stakeholders, ProPollSoil will deliver practical guidelines, tools, and policy recommendations for soil and pollinator conservation.

1.2 Related Documents

This DMP aligns with the ProPollSoil Grant Agreement, Description of the Action, Consortium Agreement, and EU Open Science and data protection policies.

1.3 Data Management Workflow and Responsibilities

The data management within ProPollSoil begins with the process of data acquisition (production) and initial recording by Team Members within Tasks and Work Packages (Figure 1). Within Tasks, Team Leaders are responsible for ensuring that their teams record data correctly and in line with the DMP guidelines and that the data is delivered to the Task Leaders. The data is then collected, compiled and processed by the Task Leaders. Task Leaders are responsible for the quality, documentation and timely deposition of datasets. Work Package Leaders are supervising Task Leaders and regularly remind Task Leaders of the guidelines set out in the DMP. Day-to-day governance in data management is overseen by the Data Management Board (DMB), comprising Astrid Neumann (TUM), Renita Danabalan (EURICE), Olaf Schmidt (UCD), Jeff Ollerton (UoS), Maria Clara Castellanos (UoS), Alexander Keller (LMU), Pierre Noiset (LMU), and Jorrit Poelen (UCSB/Ronin/GloBI). Data management is coordinated by the Technical University of Munich (project coordinator). The DMB and the project coordinator will organise three data literacy workshops, performed by Jorrit Poelen, during the course of the project (with one already held in November 2025) to ensure harmonisation of practices across partners. The content of the workshops will be adapted to the specific needs of the consortium, but may cover e.g. the performance of data reviews, registration of datasets in repositories or the usage of digital signatures.

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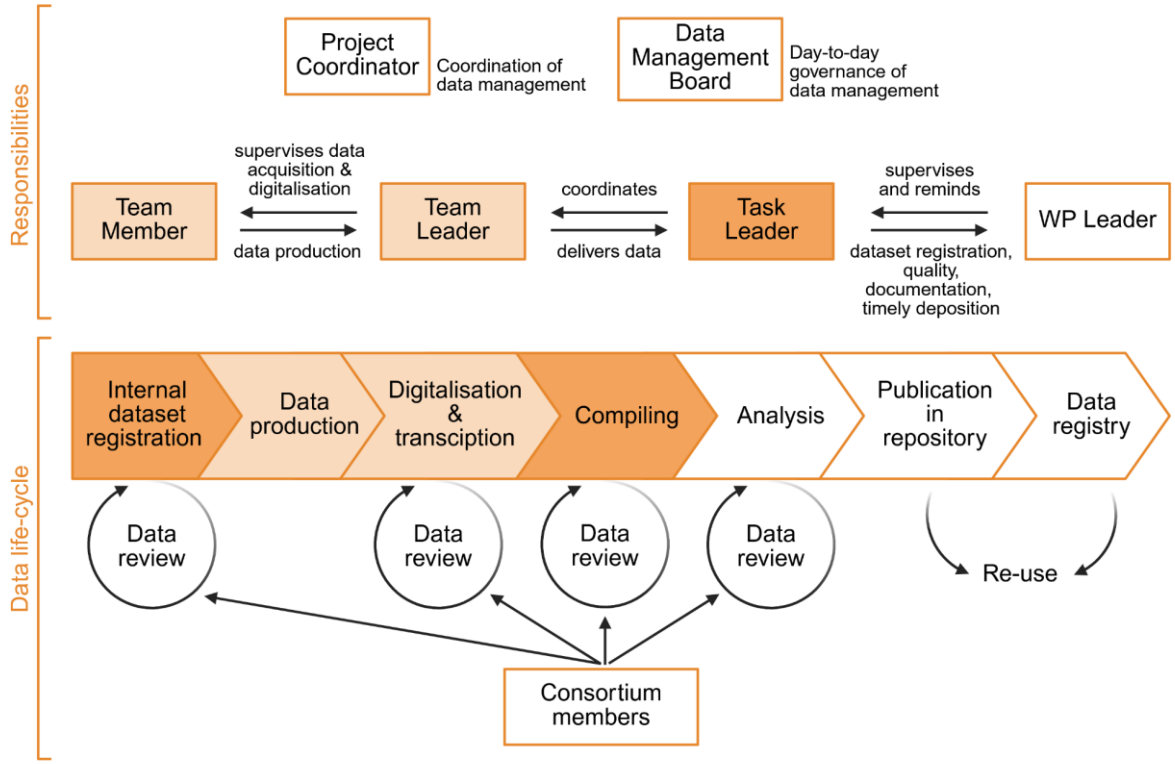


Figure 1: Data life-cycle within ProPollSoil including associated responsibilities regarding data within the consortium. Roles within the consortium (Team Member, Team Leader, Task Leader, Work Package (WP) Leader, etc.) are colour coded (light and dark orange) according to the steps in the data life-cycle. Blank shapes indicate that no fixed responsibilities have been assigned between a role and the data life-cycle, as they vary depending on Work Packages or Tasks. Created in BioRender. Neumann, A. (2026) https://BioRender.com/dcyie94.

1.4 Definitions

Darwin Core (DwC, Wieczorek et al. 2012): Darwin Core is a community-maintained biodiversity data standard that defines a shared set of terms for describing taxa and their occurrences. It enables consistent recording and exchange of information from observations, specimens, and samples across databases.

Digital Dataset: Data stored in one or more repositories and registered with one or more registries (e.g. Zenodo, GloBI (Poelen et al. 2014), EuroAPPA).

Dublin Core (DC, Kunze & Baker 2007): Dublin Core Metadata Initiative (DCMI) provides a standardized, simple set of 15 core metadata elements for describing digital or physical resources to improve information retrieval.

Ecological Metadata Language (EML, Jones et al. 2019): The Ecological Metadata Language is a community-maintained standard that provides a structured vocabulary and XML format for describing research data. Widely used in environmental sciences, it supports clear documentation, preservation, and sharing of datasets across disciplines.

Long-Term: At least five years after project end.

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GDPR (General Data Protection Regulation): The European Union's legal framework governing the collection, processing, storage, sharing, and protection of personal data.

Intermediate data: Data that have undergone some level of transformation, cleaning, formatting, or organisation, but are not yet in their final curated form. These datasets sit between raw data and final processed outputs.

Metadata: The descriptive information that explains the content, context, structure, and meaning of a dataset. It enables others to understand what the data represent, how they were created, how they should be interpreted, and how they can be reused.

Processed / final / curated data: Datasets that have been cleaned, transformed, quality-checked, harmonised, and fully prepared for analysis, publication, or archiving. These datasets incorporate all necessary documentation, metadata, and validation required for reuse.

Raw data: the original, unprocessed, and unaltered data collected directly from a source during fieldwork, laboratory analysis, surveys, third party datasets, etc and unprocessed images and videos. Raw sequencing data refers to unprocessed genomic sequences (e.g., FASTQ files).

Registry: A service indexing datasets and metadata to improve discoverability.

Repository: A digital space where data is stored, either publicly (e.g. Zenodo, DataDyrad, FigShare)) or privately (e.g. institutional servers). In this DMP, "repository" refers to trusted public repositories unless stated otherwise.

Versioning: The systematic process of tracking and managing changes made to a dataset, document, or piece of code over time. Each update results in a new, uniquely identified machine generated identifier (e.g., git commit hashes, a checksum). These identifiers can be associated with human readable version labels (e.g., v1.0, v1.1, v2.0) to further aid users in keeping track of changes over time and ensure transparency, reproducibility, and accountability.

1.5 In and Out scope

This DMP covers the management of all research data generated or reused within the ProPollSoil project, including raw, intermediate, and final curated datasets. It defines procedures for documenting, storing, reviewing, sharing, and preserving digital datasets and associated physical research materials throughout the project lifecycle. The DMP applies to all datasets (see Table 1 for types), as well as related code, protocols, and computational workflows.

The document also outlines responsibilities for data governance within the consortium, including dataset registration, review processes, metadata standards, and compliance with FAIR principles and relevant legal and ethical frameworks.

Administrative project documentation, financial reporting, and manuscripts themselves are outside the scope of this DMP. Matters related to intellectual property, access rights, and legal obligations are governed by the Grant Agreement and Consortium Agreement.

2 Data

2.1 Data Summary

ProPollSoil will generate and re-use multiple data types: observational and experimental data (soil physical, chemical and biological properties; pollinator traits and abundance, diversity and occurrence;

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plant traits; land-use and management practices); geospatial data (maps of soil properties, land use, habitats, nest positions, pollinator distributions, climatic zones); genomic and molecular data of pollinators, other invertebrates and microbes (DNA/RNA sequences, gene expression profiles); image and video data (high-resolution specimen images, monitoring footage, scans); expert survey data; citizen science data (species occurrence and nest observations); and analytical outputs including models and code. The project anticipates generating and re-using approximately 75–100 TB of data over 48 months. While most tabular datasets are relatively small (MB–GB), imaging and sequencing data dominate storage needs.

2.2 Data types

ProPollSoil will generate heterogeneous data types spanning biodiversity and survey observations, genomic sequences, ecological and interaction data, soil properties and land management information, imaging and video materials, specimen and trait records, as well as software and computational models (Table 1). A consistent format strategy will be applied across all Work Packages: data will be stored and shared in open, widely adopted, machine-readable formats wherever feasible, and proprietary formats will be avoided or accompanied by documented exports into open equivalents.

Accordingly, tabular and semi structured data will be provided in formats such as CSV and JSON, biodiversity and ecological datasets will use established exchange packages such as DwC A and EML, spatial data will use standard geospatial formats (for example GeoTIFF, GeoPackage, and shapefiles), sequence data will use community standards (FASTA and FASTQ), and image or video outputs will use widely supported archival formats (JPG, TIFF, MP4).

Code will be maintained in plain text UTF-8 encoded source form with version control, and computational model artefacts will be deposited alongside sufficient documentation and, where possible, interoperable export representations to support reuse. All prioritised data types, typical individual dataset sizes, and the corresponding target repositories and registries are summarised in Table 1.

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Table 1: Non-exhaustive list of identified trustworthy and established repositories, registries and prioritized data types in addition to general ones like Zenodo and FigShare as well as registration with EuroAPPA.

Data type	Format	Size/item	Description	Target repositories and registries
Biodiversity and Survey Data	CSV JSON XML DwC-A EML	KB-MB	Observations of species occurrences, distribution, and taxonomy.	DataDryad, GBIF, iNaturalist, Zenodo, EU Pollinator Information Hive, Edaphobase
Genomic Data	FASTA FASTQ	MB-GB	Raw sequence data and annotations.	EBI-SRA, NCBI, BOLD
Ecological Data	CSV JSON XML EML	KB-MB	Data on species interactions, habitat characteristics, and ecosystem processes.	DataDryad, EuroAPPA, GloBI, DoPI
Soil Data	CSV Shapefiles XML	KB-GB	Information on soil composition, pollution, properties, and management.	DataDryad, ESDAC, EUSO, ISRIC, FAO SOILS PORTAL, SoilWise
Land Management Data, Habitat classification	CSV Shapefiles XML	KB-GB	Information on land management practices and areas of application.	DataDryad, ESDAC, EUSO, ISRIC, FAO SOILS PORTAL, SoilWise, CORINE, Edaphobase
Monitoring Video and Image Data	JPG TIFF MP4	GB-TB	Raw pictures and video footage from field monitoring and monitoring method development. X-Ray CT of burrows.	EuroBioImaging
Specimen Images	JPG TIFF	MB-GB	High-resolution images of biological specimens.	BigBee, iNaturalist, EuroBioImaging
Specimen Traits	CSV	KB-MB	Characteristics of biological specimens observed or collected.	TRY, BigBee, OpenTraitNetwork
Code	R PY C container recipes conda yml renv	KB-MB	Code and coding environments for software, analyses & modelling.	GitHub, GitLab, Codeberg
Models	Keras h5 Rdata	KB-MB	Deep learning, mechanistic and predictive models.	GitHub, HuggingFace

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Data type	Format	Size/item	Description	Target repositories and registries
Human participant and stakeholder data (restricted)	CSV JSON PDF TXT	KB-MB	Survey and interview data, workshop and stakeholder outputs, coded transcripts. Anonymised or pseudonymised where applicable.	Controlled institutional repositories with access control; Zenodo and EuroAPPA for open metadata only
Protocols and methodological assets	PDF MD TXT CSV YAML	KB-MB	Standard operating procedures, field and lab protocols, questionnaires, sampling sheets, calibration and QA QC procedures, codebooks, annotation guidelines, and method documentation.	GitHub, GitLab, Codeberg

2.3 Data Processing, Storage, and Sharing Within the Project

2.3.1 Digitalisation and Storage of Data

The digitalisation of data will consist of two steps:

1. After every data collection event, (physical) datasheets need to be either photographed or scanned on the same day of the event. Photos or scans of datasheets will be stored on secure servers as soon as possible after the data collection event, e.g. every week.
2. Data will be transcribed (entered from datasheets into data tables) as soon as possible after a data collection event, e.g. every week, as details during the data collection may be forgotten and lost over time.

Raw data will be collected following standardised protocols and stored on at least two different locations: First, on secure institutional servers with controlled access and regular disaster recovery. Second, all project data, including raw, intermediate, and processed datasets, need to be deposited at a second secure location. For this, project partners are encouraged to deposit project data in a centralized secure Nextcloud ProPollSoil-dedicated instance hosted at LMU. In any case, each partner will be responsible for the safe storage of their data.

2.3.2 Internal Data Exchange Infrastructure

The use of a ProPollSoil-dedicated Nextcloud platform for internal data exchange is provided for all consortium partners to allow internal archiving, transparency, traceability, and coordinated data management. Access to Nextcloud is restricted to registered consortium user accounts with role based permissions. Where collaboration with external stakeholders is necessary, clearly defined individual shared folders with limited access rights will be created to ensure controlled data exchange. The Nextcloud environment provides file versioning to document changes and maintain traceability of edits. In addition, Nextcloud storage backup procedures ensure data security, including regular backup to magnetic tape systems and mirrored storage at Leibniz Supercomputing Center (LRZ) in Garching, providing redundancy and disaster recovery capacity during the active project phase.

Deposition of all final products relevant for publication, reporting, modelling, or external dissemination is mandatory within the consortium Nextcloud platform prior to external submission. This ensures internal archiving, review, quality control, and documentation of provenance. Data processing

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workflows will be documented through structured metadata, organised directory structures, and version controlled scripts.

Final curated datasets and associated code will subsequently be prepared for deposition in trusted external repositories as described in Section 3.

2.3.3 Dataset Review

ProPollSoil adopts a proactive, structured approach to ensure that all datasets produced or reused within the project are fully documented, versioned, and reusable (see Annex 1 for full details and figures). A central Dataset Registry, maintained by the DMB in collaboration with Task Leaders, will record key metadata for each dataset, including authorship, reviewers, licensing, version history, file inventories, and review status. Only registered datasets may be included in project outputs. Clear minimum requirements apply: datasets must include basic metadata (e.g. a README file), use open and transferable file formats, consist of static files that can be shared offline, and avoid reliance on proprietary or remote-only content.

Work Package Leaders are required to ensure that datasets are (1) registered early by Task Leaders (*latest* when a first version of the plan for dataset structure is decided), and that (2) periodic dataset reviews with cross-WP representation are organised and conducted by the Task Leaders (at least before each annual meeting). Review outcomes will be documented and dataset versions updated accordingly. Training and expert support (e.g. during ProPollSoil data literacy workshops) are available to facilitate this process. Together, this framework embeds data quality control and reusability into routine project practice, ensuring that ProPollSoil outputs are robust, transparent, and accessible to a wide range of stakeholders. See Annex 1 for a detailed description of the workflow, timeline and responsibilities for dataset registration and reviews within ProPollSoil.

2.3.4 Internal accessibility of data associated with manuscript during circulation

To ensure scientific quality, transparency, and internal coherence, all datasets intended for publication in association to a manuscript or external dissemination must be deposited and shared in the consortium Nextcloud at least two weeks prior to submission. Corresponding metadata, code, and documentation must be included to allow internal reproducibility checks.

The associated dataset will be shared in addition to the circulation of the manuscripts. Manuscripts will be formally announced 45 days prior to journal submission and circulated to the entire consortium at least two weeks prior to journal submission or preprint deposition. Details on how and when to share manuscripts can be found in the Management Guide (D1.1) of ProPollSoil under “Dissemination and Publication of Project Results”.

This two-week period constitutes an internal review window during which consortium members may provide comments, request clarification, or raise justified concerns related to data integrity, authorship, intellectual property rights, confidentiality, or compliance with consortium agreements. Feedback will be addressed before submission to ensure transparency, scientific quality, and alignment with project objectives.

2.4 Reuse of Data

ProPollSoil will re-use existing data from published literature, grey sources, citizen science platforms (e.g. iNaturalist), primary data provides (e.g. meteorological agencies) and established data networks including EuroAPPA, DoPI, GloBI, GBIF, SoilWise, EUSO, and SOB4ES. These datasets include soil properties, pollinator occurrence records, and time-series data from previous monitoring campaigns. Reused data will be properly referenced, harmonised, and documented to ensure provenance and compliance with FAIR principles. Aside from including data citations, contributors are encouraged to

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include original copies of reused datasets in their data publications to facilitate review of data products and avoid unnecessary dependencies on third party data repositories.

2.5 Purpose of Data Generation in Relation to Objectives

Existing datasets will establish baselines, enable cross regional comparison, identify knowledge gaps, and facilitate analysis in relation to, for example, soil taxonomy and climatic zones. Newly generated data from field campaigns, laboratory experiments, Case Study Areas (CSAs), expert survey and citizen science initiatives will provide systematic, harmonised, high-resolution information on soil properties, pollinator diversity, plant material properties, species interactions, and management practices under real world conditions.

Data generation will adhere to recognised domain-specific community standards to ensure comparability and long-term interoperability. For example, taxonomic identification will rely on established molecular markers such as COI for pollinators, 16S rRNA for bacteria, and ITS for fungi, aligned with reference frameworks such as Barcode of Life Data System or NCBI. Soil sampling and analytical procedures will follow established European protocols, including those used in LUCAS Soil where applicable, to ensure methodological compatibility with continental monitoring efforts.

The integration of standardised, existing and newly generated datasets will support hypothesis testing, predictive modelling, spatial mapping, and the development of decision support tools. These outputs will underpin evidence based recommendations for sustainable soil management, pollinator conservation, and agroecological transition.

2.6 Usefulness of Data Outside the Project

The data generated and curated within ProPollSoil will be relevant beyond the consortium and project lifetime. They will support researchers in soil science, biodiversity, and agroecology; inform policy development and monitoring frameworks; guide land managers and farmers in sustainable practice; assist industry actors in environmental assessment; and strengthen citizen science and NGO initiatives focused on soil health and biodiversity conservation.

By ensuring standardisation, open accessibility where possible, and interoperability with European infrastructures, the datasets will contribute to long-term environmental monitoring, policy evaluation, and cross project synthesis at national and European scales.

3 FAIR Data Management

3.1 FAIR Data Summary

All ProPollSoil data will follow FAIR principles to maximise findability, accessibility, interoperability, and reuse. Datasets will receive persistent identifiers (e.g. DOI, ARK) on depositing at least two identical copies across independent research data infrastructures, one of them being Zenodo and one other recognised domain specific repositories e.g. DataOne, DataDryad, FigShare). Machine readable metadata following community standards will ensure discoverability and structured documentation. Datasets will also be indexed in relevant disciplinary registries. Data will be openly available under permissive licences wherever possible. Metadata will always be published under CC0. Any restrictions due to legal, ethical, or contractual constraints will be transparently documented and managed by the Data Management Board, with metadata remaining publicly accessible. Interoperability will be ensured through the use of standardised vocabularies, ontologies, and open formats. Reuse will be supported through detailed documentation, version control, linked persistent identifiers, and open sharing of code and computational workflows. Data provenance will be documented through structured metadata, version histories, and explicit workflow descriptions linking raw data, processed

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data, and derived products. Relationships between datasets, scripts, and publications will be expressed using persistent identifiers to ensure traceability.

3.2 Making Data Findable

3.2.1 Identifiers and Repositories

All datasets will be assigned persistent identifiers upon public deposition, typically DOIs or ARKs, to ensure citability, traceability, and long-term accessibility. Versioning will be implemented where datasets are updated, with clear documentation of changes between releases. The primary repository will be Zenodo, organised through a dedicated ProPollSoil community to ensure thematic aggregation, consistent curation, and project level visibility. Additional disciplinary repositories will be used where appropriate, including GBIF for biodiversity data, ENA for genomic data, and soil repositories catalogued by SoilWise to ensure visibility within the EU Soil Mission. Repository selection will follow the principles of disciplinary best practice, FAIR compliance, long term sustainability, and international visibility. An overview of all suggested repositories, in addition to Zenodo, and corresponding data types is provided in Table 1.

3.2.2 Metadata Standards

Comprehensive, machine readable metadata will accompany all datasets to facilitate interpretability, interoperability, and long term reuse. Metadata records will include, at minimum, dataset title, authors and affiliations, persistent identifier (DOI, ARK), abstract, methodological description, sampling design, variable definitions, units, data types, file formats, software dependencies including version numbers, processing workflows, quality control procedures, and explicit assumptions or data limitations. Additionally, we encourage the consortium to provide a digital signature (e.g. hashes) in the metadata (Elliott et al. 2023).

For all produced data (e.g. lab measurements, field observations, derived data records) minimum mandatory metadata elements will include sampling date, geographic location with coordinate reference system, site or habitat identifier, sampling protocol reference, and globally unique observation and observer identifiers.

Community accepted standards will be applied according to data type:

- Dublin Core (DC) for general resource description and repository level interoperability
- Darwin Core (DwC) for species occurrence and taxonomic information
- Minimum information about a marker gene sequence (MiMarks, Yilmaz et al., 2011) for microbiome descriptors
- Ecological Metadata Language (EML) for ecological datasets and sampling metadata
- ISO 19115-1:2014/Amd 2:2020 and related ISO standards for geospatial and soil data

3.2.3 Specific Considerations

3.2.3.1 Soil data

Soil datasets will follow relevant analytical and methodological standards, e.g. ISO 11277:2020/Amd 1:2024 for particle size distribution and ISO 10390:2021 for soil pH determination. In addition to default metadata, metadata will also document sampling depth, horizon designation where relevant, laboratory methods, calibration procedures, and analytical uncertainty.

3.2.3.2 Pollinators

Pollinator datasets will implement Darwin Core vocabulary and recognised taxonomic reference frameworks. Metadata will explicitly describe life stage of organisms, sampling effort, trapping or

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observation protocols, temporal coverage, spatial resolution, sampling coordinates, and identification procedures, including molecular workflows where applicable.

3.2.3.3 Human Participants

Expert survey, socioeconomic and stakeholder related datasets will be anonymised or pseudonymised prior to deposition and handled in full compliance with the GDPR. Metadata will clearly define access conditions, consent frameworks, data processing purposes, and any restrictions on reuse. Sensitive data will be deposited in controlled access repositories where required.

3.2.4 Registries and Indexing

To maximise discoverability for re-use, datasets will be registered and indexed in domain-specific research data registries and aggregators (see Table 1). All datasets will be registered on a dedicated ProPollSoil dataset registration in EuroAPPA. Additionally, individual datasets will be indexed through specific infrastructures where applicable, for example: species interactions through Global Biotic Interactions (GloBI; organism trait data through Open Trait Network; soil and land management through European soil data infrastructures including the European Soil Data Centre and its European Soil Database (ESDB), European Soil Observatory (EUSO) and its Soil Health Dashboard, and SoilWise, as well as global platforms such as ISRIC World Soil Information and the FAO Soils Portal.

3.3 Making Data Accessible

Most data and associated metadata will be made openly available upon publication under permissive licences such as CC BY or CC0, depending on data type and community practice and if not in conflict with IP or IPR. Licensing terms will be clearly specified at the point of deposition to ensure legal clarity and facilitate reuse.

Certain datasets may require restricted access due to legal, ethical, privacy, intellectual property rights, or contractual constraints, for example personal data, sensitive socioeconomic information, or proprietary industry contributions. In such cases, restriction needs to be approved through formal request procedures overseen by the Data Management Board. Decisions will be documented and made public alongside metadata to ensure accountability and consistency. The metadata will also document how access can be gained in GDPR compliant ways. Requests must include institutional affiliation, contact details, intended use, and agreement to clearly defined terms of use. All handling of personal data will comply with applicable European data protection legislation.

Metadata describing all datasets, including those under restricted access, will remain openly accessible (CC0) to ensure discoverability and transparency. Persistent identifiers and landing pages will remain active regardless of access status.

Embargo periods may be applied to allow for scientific publication, intellectual property assessment, or coordination with stakeholders. Embargoes will not exceed five years after data generation, unless justified by legal or contractual requirements. Any exceptions will require formal approval by the Data Management Board and written documentation of the rationale. Upon expiry of the embargo, datasets will be made openly accessible without additional restriction.

3.4 Making Data Interoperable

Interoperability will be ensured through the systematic use of community endorsed data standards, controlled vocabularies, and formal ontologies appropriate to each data type. Biodiversity and pollinator occurrence data will apply Darwin Core (DwC) terms. Ecological datasets will use Ecological Metadata Language (EML) for structured sampling and methodological descriptions. Soil and geospatial data will follow relevant ISO standards alongside recognised soil analytical standards. Semantic relationships will be described using established ontologies such as the OBO Foundry framework, including the Relation Ontology where appropriate. Controlled vocabularies and

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recognised taxonomic and geographic reference systems will be used to harmonise terminology across datasets.

Data will be stored in open, non-proprietary, and widely adopted formats, including CSV, JSON, XML, DwC A, EML compliant XML, FASTA, FASTQ, GeoTIFF, and shapefiles, to facilitate machine readability and long term accessibility (Table 1). Where binary formats are necessary for modelling outputs, accompanying documentation and, where feasible, interoperable export formats will be provided.

Qualified references will be implemented using persistent identifiers to explicitly link related datasets, publications, software, and models. Relationships will be expressed using meaningful, standardised predicates such as “is derived from”, “is part of”, “is documented by”, or “influences”, ensuring semantic clarity and enabling machine actionable integration across knowledge graphs and repositories.

3.5 Increasing Data Reuse

Each dataset will be accompanied by comprehensive documentation to ensure interpretability and long-term reuse. This will include structured README files describing study design, sampling protocols, methodological workflows, variable definitions, units, coding schemes, data cleaning procedures, and known limitations. Where applicable, formal codebooks will define categorical variables, controlled vocabularies, and data transformations. Relationships between datasets, scripts, and publications will be expressed using persistent identifiers to ensure traceability.

All analysis scripts, workflows, and computational environments (R, Python, Bash) will be shared via GitHub or Codeberg, with stable archived releases deposited in Zenodo to obtain persistent identifiers. Where relevant, containerised or environment specification files (for example R or conda environments, or Docker definitions) will be provided to facilitate computational reproducibility.

Licensing terms will explicitly permit reuse, redistribution, and adaptation, subject to attribution requirements where applicable. Any reuse constraints, including ethical or contractual limitations, will be clearly documented in metadata and access statements.

4 Other Research Outputs

4.1 Physical samples

ProPollSoil will generate physical research outputs including soil samples, pollinator specimens, plant material, and associated biological material collected during field campaigns and controlled experiments. These materials constitute valuable long-term scientific assets beyond the lifetime of the project, and a best effort should be made to deposit these assets into long-term institutional collections to facilitate reuse beyond this project where possible. All samples will be assigned unique, persistent internal identifiers linked to digital records in the project database. Associated metadata will include geographic coordinates, site identifiers, sampling date, collector, institutional affiliation, sampling protocol, environmental conditions, and any subsequent processing steps. Where feasible, identifiers will be cross-referenced with dataset persistent identifiers (DOI or ARK) and publications to ensure traceability between physical and digital outputs.

Soil samples will be air dried, sieved where appropriate, labelled with durable identifiers, and stored in institutional facilities under controlled temperature and humidity conditions. Storage will ensure physical integrity and traceability for at least ten years after project completion. Methods for pretreatment will follow ISO 11464:2006 and methods for long-term storage will follow ISO 18512:2007.

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Pollinator specimens and biological material will be preserved according to established entomological standards, including pinning in archival quality cases or preservation in ethanol of appropriate concentration. Where possible, specimens will be deposited in recognised institutional collections or museums to ensure long term curation, professional conservation, and scientific accessibility.

Movement of biological samples across borders within Europe will comply with relevant national and EU regulations, including phytosanitary and biodiversity protection laws, and necessary permits will be obtained in advance to avoid delays. Where export/import restrictions apply, samples will either remain in-country for analysis or be processed into non-restricted derivatives before shipment.

A publicly accessible catalogue of physical outputs will be maintained, including basic metadata and designated contact persons for access requests. Access and reuse will be governed case by case through material transfer agreements or institutional policies, in compliance with legal, ethical, and institutional requirements. Long term preservation responsibility will rest with the hosting institutions or accredited collections.

4.2 Other Scientific and Knowledge Outputs

The project will generate a range of scientific and knowledge outputs, including peer-reviewed publications, data papers, preprints, conference presentations, posters, software releases, methodological protocols, and policy relevant briefs. The Data Management Board strongly encourages the use of Open Access dissemination formats to maximise transparency, early visibility, and long-term reuse of project results. While not formally binding, consortium members are expected to actively consider preprint deposition, publication of data papers for major datasets, and open archiving of conference, workshop and consortium meeting materials wherever policies and contractual obligations permit.

Preprints will be deposited in recognised open repositories (e.g. https://arxiv.org/, https://ecoevorxiv.org/) to accelerate dissemination and allow community feedback prior to formal publication. Data papers will be used, where appropriate, to provide structured, citable documentation of substantial datasets. Posters, presentations, and workshop materials will be archived in the ProPollSoil Zenodo community to ensure continued accessibility beyond the event itself.

All such outputs will, where feasible, be linked to underlying datasets, software, and models via persistent identifiers to facilitate traceability and reproducibility. This integrated approach strengthens scientific impact, supports knowledge transfer to stakeholders and policy frameworks, and enhances long term reuse of project results.

4.3 Modelling and Socioeconomic Research Outputs

In addition to physical samples and scientific publications, ProPollSoil will generate socioeconomic datasets and modelling-related research outputs. These may include anonymised survey datasets, stakeholder workshop summaries, aggregated socioeconomic indicators, model input datasets, model calibration files, scripts, and scenario outputs generated through ecological, biophysical and socioeconomic modelling frameworks.

Wherever possible, modelling inputs will rely on aggregated or anonymised data to ensure compliance with data protection regulations. Model outputs and derived indicators will not contain identifiable personal information.

Version-controlled repositories will be used for modelling scripts and workflows to ensure transparency and reproducibility. Final model code and documentation will be deposited in trusted repositories (e.g. Zenodo or institutional repositories) and linked to associated datasets and publications through persistent identifiers.

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Where legal and ethical constraints permit, selected modelling datasets and scenario outputs will be made openly available to support reuse, validation and policy uptake. Access to restricted modelling inputs will be managed according to consortium agreements and applicable data protection rules.

This integrated approach ensures traceability between physical samples, datasets, modelling workflows and scientific outputs, thereby strengthening reproducibility, long-term reuse and policy relevance of project results.

5 Allocation of Resources

Costs for FAIR data management in ProPollSoil will primarily involve personnel time for data curation, metadata creation, and quality assurance, as well as infrastructure for storage and long-term preservation. Direct costs include maintaining secure institutional servers, the centralized and ProPollSoil-dedicated Nextcloud instance, backups, and minor hardware upgrades. Indirect costs include staff time for preparing interoperable datasets, assigning DOIs, and uploading to trusted repositories. At time of writing, Zenodo does not charge for depositing (data) publications. Also, registries such GBIF and GloBI do not charge for adding datasets to their registries at this point in time. Actively managed datasets such as iNaturalist and DoPI are provided at no cost currently. Costs for managing physical outputs (soil and pollinator samples) will be covered by institutional facilities, with minor expenses for cataloguing and compliance with cross-border regulations (permits and shipping). Overall, FAIR implementation costs are integrated into the project budget under data management and dissemination activities.

Another type of cost will be the article processing charges (APC) for publishing data in open-access journals. Many of ProPollSoil partners have institutional or national Read & Publish agreements with publishers that cover open access (OA) charges and/or OA funds and will utilise these agreements and funds where possible. Each partner is expected to use their budget responsibly and prioritise open access publications. In case of uncleared costs by mutually generated data, the responsibility and costs will be discussed in the Executive Board (WP Leaders and PMT) and/or General Assembly meetings.

6 Data Security

Raw, intermediate, and final datasets, together with associated metadata, will be stored on institutional servers at the respective partner organisations in accordance with their established data preservation and security policies. Access to local storage systems will be restricted through password protected user accounts and role-based permissions. Data integrity will be facilitated through automatic and regular backup procedures implemented at each partner's institution. Sensitive data will be anonymised or pseudonymised where required and stored in restricted access folders in compliance with applicable European and national data protection regulations.

In addition, a dedicated Nextcloud instance hosted at LMU is provided as the central internal collaboration platform for ProPollSoil. Deposition of all final datasets and associated products relevant for publication, reporting, modelling, or external dissemination is mandatory in this platform prior to submission or external release. This ensures internal archiving, quality control, and documentation of provenance. Deposition of raw and intermediate data in Nextcloud is strongly encouraged to facilitate internal archival, backup, transparency, traceability, and coordinated data management, but remains the responsibility of individual partners according to their institutional policies.

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Final curated datasets and associated code will subsequently be deposited in trusted external repositories as described in Section 3 to ensure long term preservation and public accessibility where appropriate.

7 Ethics

7.1 Biological and Ecological Data

All collection, handling and transfer of insects and other biological materials will comply with national and international regulations. Required permits (including those related to protected species and cross-border transfer of biological materials) will be obtained prior to sampling and in agreement with national authorities. Shipping of live soil will adhere to relevant legislation including the Plant Health Law (Regulation EU 2016/2031).

Genetic resources and any traditional knowledge associated with them will be accessed, stored and handled in accordance with applicable Access and Benefit-Sharing (ABS) legislation (e.g. Nagoya Protocol) and relevant national regulatory requirements. Where applicable, benefits arising from the use of genetic resources will be shared fairly and equitably under mutually agreed terms.

Such regulatory constraints may limit open sharing of certain raw biological datasets. However, aggregated, anonymised or derived datasets that do not compromise regulatory compliance will be made openly available through appropriate repositories in line with the project's open science commitments.

7.2 Human Participant Data

All data collected from human participants will follow the principles of informed consent, confidentiality, data minimisation and purpose limitation. Consent procedures will explicitly include permission for anonymised data sharing among project partners, long-term preservation of anonymised datasets, and use of collected information for research and modelling purposes.

Personal data (e.g. contact details for follow-up communication or workshop organisation) may be collected through interviews, surveys, workshops or stakeholder consultations. Appropriate safeguards will be implemented to ensure protection of participants' personal data in accordance with the General Data Protection Regulation (GDPR) and provisions of the Consortium Agreement.

No special categories of personal data (e.g. ethnicity, health data, political opinions, religious beliefs or criminal records) will be collected or processed unless explicitly required and ethically approved. No pre-existing personal datasets will be merged with newly collected identifiable data. Participants will retain the right to withdraw their identifiable data at any time prior to anonymisation.

A key component of the project involves the use of collected data within modelling frameworks (e.g. ecological, biophysical, socioeconomic and bio-economic modelling chains). Research protocols and modelling methodologies will adopt an inclusive and ethically responsible approach by:

- Ensuring that model design, calibration and interpretation avoid bias related to gender, ethnicity, income level, education, age or other socioeconomic characteristics;
- Preventing under-representation of minority or marginalised groups in datasets used for modelling and decision-support tools;
- Using aggregated or anonymised data in modelling activities wherever possible;

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- Ensuring transparency in assumptions, parameters and data transformations used in modelling processes;
- Validating model outputs through expert review and stakeholder consultation to minimise risks of misinterpretation.

Data used for modelling will be retained only for the duration necessary for research, reporting and potential reuse in line with open science principles. Unless otherwise agreed for long-term open access, identifiable or sensitive datasets will be deleted or securely archived no later than five (5) years after the end of the project, in accordance with GDPR and institutional data retention policies. Any changes to retention periods will be specified in updates of the Data Management Plan.

All requests to reuse data for purposes other than those initially defined will be handled transparently, fairly and in compliance with applicable data protection rules. Data shared publicly will be fully anonymised or aggregated.

7.3 Ethical issues related to Artificial Intelligence

AI tools (e.g., ChatGPT, Grammarly) are used responsibly and ethically throughout the project, with strict adherence to the living guidelines on the responsible use of generative AI in research (European commission, 2025). Datasets will be transparently labelled as "AI-processed" and accompanied by metadata. Publications that use AI for data processing will include information about their application and associated risks. Deep-learning systems will follow the Assessment List for Trustworthy Artificial Intelligence (ALTAI) for mandatory self-assessments, comprehensive risk assessment and pending approval by the Steering Committee, with accountability remaining with the developers. We will use open-source AI frameworks and publish our code on GitHub to enhance transparency, to foster external review and to allow understanding of processes.

7.4 Data Protection, Storage and Retention

Appropriate safeguards for protecting personal data will include:

- Application of the principle of data minimisation;
- Anonymisation or pseudonymisation where required;
- Secure storage and controlled access to datasets;
- Training and guidance for researchers on data protection and ethical handling of data.

Processing and handling of personal data will follow the provisions of the Grant Agreement and applicable EU regulations. Personal data will not be transferred to the European Commission unless strictly required; in such a case, all relevant procedures and privacy statements will be followed.

All stakeholder engagement, data collection, modelling activities, dissemination and communication actions will be conducted in full compliance with GDPR and recognised ethical standards for research involving human participants and environmental data.

8 Data Ownership and Intellectual Property

Data ownership and access rights follow the Consortium Agreement (paragraph 8) and Horizon Europe Grant Agreement (Annex 5 - Article 16), balancing open science with legitimate interests and IP protection. These terms were designed to uphold a fair, and transparent approach, supporting open collaboration and responsible dissemination of results.

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1. Each partner retains ownership of the results/data they individually produce.
2. In the case of joint production, ownership shall be shared.
3. Data will be made openly available (CC-BY or CC0 licence) if not in conflict with IP or IPR.

ProPollSoil will follow an open approach for sharing and obtaining knowledge. Described at the proposal stage, ProPollSoil will combine scientific and technical know-how and background IP from the individual consortium members to develop new innovations in the field of soil and pollinator health and biodiversity. An internal IP manual provides a structured framework for identifying, protecting, and exploiting intellectual assets arising from the project and IP rights will be periodically reviewed and adapted depending on the needs of the project.

9 Conclusions

This ProPollSoil Data Management Plan offers guidance on how data and associated physical samples are acquired, processed, stored, reviewed, and disseminated. Data contributions by consortium members are shared and reviewed in the ProPollSoil Nextcloud. The ProPollSoil Data Management Board facilitates data reuse through a collaborative and iterative data review process throughout the lifetime of the project. The ProPollSoil Data Management Plan is a living document and will change during the course of the project according to the needs and inputs of the consortium.

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11 Disclaimer

ChatGPT (version 5.2) was used to refine structure and individual paragraphs of this DMP.

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12 Annexes

Annex 1: Data Registration and Review Process

A pro-active approach is taken to facilitate the production and publication of re-usable datasets. This is done through registration of *all* ProPollSoil datasets in the ProPollSoil Dataset Registry ("the data registry"). Only registered datasets may be included in ProPollSoil products (e.g. manuscripts, data papers). In addition, dataset authors are expected to conduct data reviews to assess whether the dataset is fit for use. To help facilitate this data review process (see section “Review Process” below), the Data Management Board will organize data literacy workshops, provide an internal repository and a shared dataset registry (e.g. within ProPollSoil Nextcloud), as well as offer access to data publication experts (e.g. members of the DMB).

The entities related to the dataset registration and review process can be seen in Figure S1.

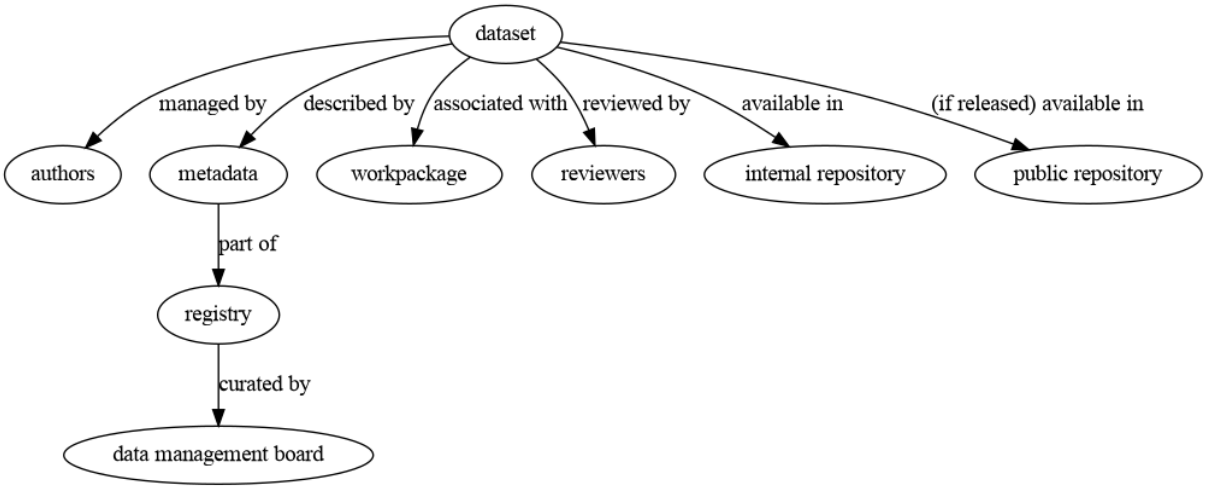


Figure S1: An entity-relationship model showing the relationships (e.g. part of, curated by, available in) between entities (e.g. authors, datasets, registry, reviewers, internal repository) involved in the dataset registration and review process.

Dataset Registry

To keep track of the dataset reviews, the Data Management Board works with Task Leaders to create and maintain a registry of *all* ProPollSoil datasets. This internal registry is available through consortium supported infrastructure (e.g. ProPollSoil Nextcloud). Registering datasets early enables a timely review of their structure, which is one of the most valuable steps in preparing a high-quality dataset. Early review helps to identify and correct errors, while also promoting consistency across all project datasets. Datasets will therefore be registered as early as possible but *latest* as soon as a first version of the plan for dataset structure is decided (e.g. a spreadsheet to be used in field data collection).

To register a dataset, a first version of the following information is provided by the Task Leaders to the Data Management Board:

1. Title (mandatory)
2. Authors and contact information (mandatory)
3. Abstract (mandatory)
4. Bill of materials - an exhaustive list of files (name, size, digital signature, location) that make up the dataset (when available)
5. Associated Work Package(s) producing the data (mandatory)
6. List of review notes (when available)

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7. Status: only dataset marked as “published” can be shared publicly. Dataset in status “draft” or “review” need consent of authors before use (mandatory)
8. License: license under which the dataset will be distributed: CC0 (preferred) or CC-BY. If other licenses are desired, the Data Management Board needs to be consulted (mandatory)
9. Version history (mandatory for publicly released versions)

For example, consider the following registration entry:

```
title: A Checklist of European Insect Pollinator Families
authors:
   - Claus Rasmussen
   - Maria Clara Castellanos
status: draft
license: CC0
date: 2026-02-22
version: 0.2
workpackage: WP2 - Existing knowledge
abstract: Insects play an important economical role in pollinating
flowering plants. To help gather existing knowledge on these pollinators,
this taxonomic checklist of European insect pollinator families was
created.
files:
  - name: README.md
    size: 1kB
    digital signature: hash://md5/05a561607d2710339191d6c95169d6e3
  - name: pollinators.csv
    size: 1.1MiB
    digital signature: hash://md5/499c18fac4cc24ea272e7f7cae591cc5
history:
  - version: 0.1
    date: 2026-01-15
    notes: first draft of publication with skeleton data tables
```

Dataset Requirements

To help guide the dataset review process, the minimal requirements for ProPollSoil datasets are:

1. **Minimal metadata:** Dataset includes a UTF-8 encoded “README.md” or “README” text file including **title**, **authors**, **abstract**, **publication date** and an exhaustive list of **files included** in the dataset (including the README).

2. **Open data formats:** If a dataset includes data records in proprietary (e.g. xlsx, docx, pptx) or binary (e.g. RDS, RData) file formats, equivalent text-based versions are also provided (e.g. csv, txt, markdown). Raster data such as images or geospatial maps are provided in open binary formats (e.g. png, GeoTIFF, GeoPackages).

3. **Transferable:** Allows for transferring an unaltered copy of the versioned dataset via a thumbdrive or other external storage medium. For instance, a specific copy of a text file "data.txt" can be transferred onto a thumbdrive as is, whereas a google doc cannot: a google doc needs to be exported in order to be transferred. In short, datasets must consist of static files and should not rely on remote content (e.g. remote images, dynamic content).

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4. **Offline enabled:** Dataset can be used offline (e.g. once retrieved, no internet connectivity nor external authentication methods are needed to access the dataset).

These minimal requirements can be updated in later versions of the Data Management Plan, and Work Package Members are encouraged to share their feedback with the Data Management Board.

Review Process

Task Leaders are expected to schedule dataset reviews as part of their Work Package activities. Work Package Leaders are responsible for ensuring that Task Leaders organize and conduct data reviews. These data reviews include the following steps:

1. **Add/update registration:** Verify that the dataset under review has a complete, accurate, and up-to-date, entry in the data registry.

2. **Schedule a review:** Work Package Leader ensures that data reviews are organized at least *before each annual meeting*. Multiple datasets may be discussed in a single data review. In addition to the author of the dataset (e.g. the data producer), a data review should involve at least one member from another Work Package, who is not connected to the preparation of the dataset.

3. **Before a review:** Prior to a data review, a specific version of the dataset is selected and made available for review. If no data yet exists, placeholder data is provided to represent the expected data records or binary files.

4. **After a review:** After a data review, meeting notes are shared with data review participants. In addition, the dataset entry in the data registry is updated to include a reference to these notes. At the minimum, the meeting notes include: The date, attendees, version of dataset under review, and follow-up action items.

Figure S2 summarizes the process visually.

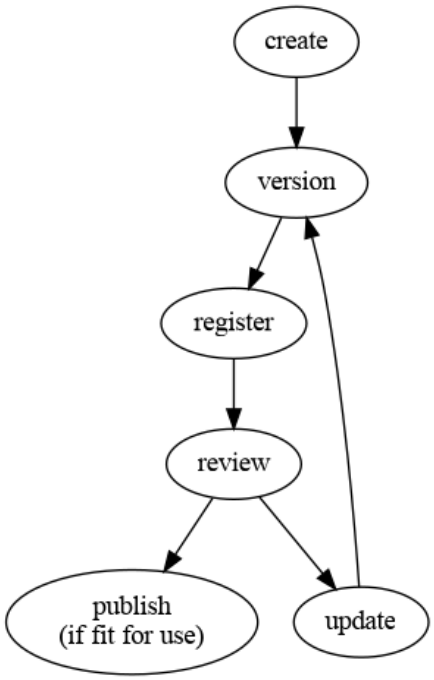


Figure S2: A process diagram showing the steps in the dataset registration and review process: First, a dataset is created and a specific version is registered with the data registry. If the dataset version is determined to be fit

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for use in a review, the data may be published in an external repository. After review, the dataset may be updated to start another version, register, and review cycle.

The review notes are important documents that indicate that the Work Package is making an effort to produce reusable data products. An (minimal) example of review notes is shown below.

```
Review date: 2026-02-23

 Attendees: Claus Rasmussen, Jorrit Poelen, Nick Balfour, Jeff Ollerton

 Dataset under review: European checklist of pollinators v0.1 including
README, and pollinators.csv.

 Action items:
   - turn dataset into a valid Darwin Core Archive by adding meta.xml and
eml.xml files.
```

Note that review notes are not shared publicly, unless the review participants agree to do so. This is to encourage open discussion on how to produce reusable data products in a pragmatic way. As the authors become more comfortable sharing their dataset, they may choose to involve external reviewers prior to publication so that external stakeholders can participate in the review process prior to publication.

Dataset Review Support

Resources are available to help support data review. These include, but are not limited to, ProPollSoil data literacy workshops, and access to data integration and publication experts.

Dataset Review Summary

As mentioned in the proposal (ProPollSoil Consortium, 2025) as well as project kick-off presentations (Poelen, 2025a, b), publishing reusable datasets is a group effort and requires careful planning. The data review process outlined above is designed to facilitate the production of reusable datasets. Work Package Leaders, Task Leaders and Task Members are encouraged to conduct data reviews early and often to make sure that their data products can be used by the many actors (e.g. researchers, farmers, industry, NGOs, citizens and municipalities) that ProPollSoil engages with.

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